2.2.6 随时间变化的电场. 电极化漂移.

$\frac{\partial E}{\partial t} \neq 0$., $(\partial_x + \partial_y + \partial_z)E = 0$. $B = const$.

$$\vec{B} = B\hat{z} \qquad \vec{E} = E_0 e^{i\omega t}\,\hat{x}$$

$$m\frac{d\vec{v}}{dt} = q[\vec{E} + \vec{v}\times\vec{B}].$$

$$\therefore \begin{cases} \dot{v}_x = \frac{qB}{m}v_y + \frac{q}{m}E = \pm\omega_c v_y \pm \omega_c \frac{E}{B}. \\ \dot{v}_y = -\frac{qB}{m}v_x = \mp\omega_c v_x. \end{cases}$$

$$\Rightarrow \begin{cases} \ddot{v}_x = -\omega_c^2 v_x = \pm\omega_c \frac{\dot{E}}{B} \qquad (\dot{E} = i\omega E). \\ \ddot{v}_y = -\omega_c^2 v_y - \omega_c^2 \frac{E}{B}. \end{cases}$$

$$\therefore \begin{cases} \ddot{v}_x = -\omega_c^2\left(v_x \mp \frac{i\omega}{\omega_c}\frac{E}{B}\right) \\ \ddot{v}_y = -\omega_c^2\left(v_y + \frac{E}{B}\right). \end{cases}$$

$$v_p \equiv \pm\frac{i\omega E}{B}, \qquad v_E \equiv -\frac{E}{B}.$$

$$\therefore \begin{cases} \ddot{v}_x = -\omega_c^2 v_x + \omega_c^2 v_p \\ \ddot{v}_y = -\omega_c^2 v_y + \omega_c^2 v_E. \end{cases}$$

假设解形式为：

$$\begin{cases} v_x = v_\perp e^{i\omega_c t} + v_p \\ v_y = \pm i v_\perp e^{i\omega_c t} + v_E \end{cases}$$

$$\ddot{v}_x = v_\perp e^{i\omega_c t}(i\omega_c)^2 \pm \frac{i\omega E}{\omega_c B}(i\omega)^2$$

$$= v_\perp e^{i\omega_c t}(-\omega_c^2) + \frac{i\omega E}{\omega_c B}\cdot\omega_c^2$$

$$\Rightarrow \omega_c^2 \pm \omega^2 = 0.$$

$$\ddot{v}_y = -\omega_c^2 v_y + (\omega_c^2 - \omega^2) v_E.$$

$\omega_c^2 - \omega^2 \longrightarrow \omega_c^2$, $\omega^2 \ll \omega_c^2$

(1) v_E : $\vec{E} \times \vec{B}$

(2)